

ECON 101—Introduction to Microeconomics
Spring 2026

FINAL EXAM

April 30, 2026

General Instructions

- The exam has a duration of **110 minutes**. If you finish early, you may leave within the first 100 minutes only. Otherwise remain quietly until the exam concludes.
- All answers must be written in the answer sheet. Scratch work will not be considered. You may not take this booklet with you.
- **Allowed on your desk or person:** ID, writing instruments, ruler, and a simple calculator. **Nothing else.** No extra paper, no graphing calculators, no phones, no smart devices, no electronics, no unauthorized materials of any kind. Any item not explicitly listed above must be placed outside the room or at the front before the exam begins. No exceptions.
- You may request a bathroom break, but only one person may be out at a time.
- Crying is allowed, but quietly, and not on the exam paper.
- Good luck and may your choice of inputs maximize your profits!

Exam Instructions and Academic Dishonesty Policy

Before the exam begins:

- **Allowed on your desk or person:** ID, writing instruments, ruler, and a simple calculator. **If it is not on this list, it is not allowed.** All prohibited items must be placed outside the room or at the front before the exam begins.
- Wait until everyone is seated. Do **not** write anything—including your name and ID—until you are explicitly told to do so.

At the end of the exam:

- When time is called, put all writing instruments down immediately, close your booklet, keep your hands away from it, and remain seated until dismissed.

Academic Dishonesty:

- Any form of cheating, including any deviation from the instructions above, will be met with the **harshest consequences available**. There will be **no sympathy and no leniency**.

Answer Sheet Instructions

- **Record your exam version on the answer sheet.** If no version is indicated, the lowest-scoring version will be applied. No exceptions.
- **Use only the provided pencil and erase cleanly if you make a mistake.**
- This exam has 40 questions worth a total of 100 points. Each question is worth 2.5 points.

Exam Version 3

Consider a firm with the following production function:

$$Q = A \cdot L^{0.3} K^{0.3} M^{0.2}$$

where:

- Q is the quantity of output,
- A is total factor productivity,
- L is the quantity of labor input,
- K is the quantity of capital input,
- M is the quantity of material input.

1. Suppose the firm increases all inputs by a constant factor $t > 1$. Which of the following statements correctly describes the returns to scale of this production function?

- (a) The production function exhibits **increasing returns to scale**.
- (b) The production function exhibits **variable returns to scale** depending on the value of t .
- (c) The production function exhibits **decreasing returns to scale**.
- (d) The production function exhibits **constant returns to scale**.

2. In the short run, which of the following is true?

- (a) The firm can freely enter or exit the market.
- (b) At least one input is fixed.
- (c) All inputs are variable.
- (d) All costs are sunk.

3. Anel is a baker at “Kink 101” bakery. She’s preparing a new glaze recipe and came up with the following:

$$g_A = \min\{s, 4c\},$$

where s are units of sugar, c are units of cream, and g_A are units of glaze according to Anel’s recipe. To produce 3 units of glaze, what is the minimum amount of cream Anel needs?

- (a) $\frac{3}{4}$
- (b) 3
- (c) 1
- (d) $\frac{1}{4}$

4. Malika also works at the bakery, but she thinks Anel’s glaze is too sweet, and that the perfect glaze recipe should be:

$$g_M = \min\{6s, c\},$$

where g_M are units of glaze according to Malika’s recipe. What is the minimum amount of sugar Malika needs to produce 3 units of glaze?

- (a) $\frac{1}{2}$
- (b) 3
- (c) 6
- (d) $\frac{3}{4}$

5. Medina, who is the owner of Kink 101 bakery, has to decide which glaze recipe will be used for all cakes. From her point of view, in the production of cakes, c , both recipes are perfect substitutes, e.g., $c = g_A + g_M$. Suppose that sugar costs 1 per unit and cream costs 10 per unit. Which recipe will Medina choose if she wants to minimize cost?
- (a) Malika's recipe
 - (b) Medina is indifferent
 - (c) Anel's recipe
 - (d) Cannot be determined
6. Which of the following statements is correct?
- (a) Variable costs cannot be avoided by shutting down.
 - (b) In the long run, there are no sunk costs, but there may be fixed costs.
 - (c) In the long run, all costs are sunk.
 - (d) Sunk costs vary with the level of output.
7. A profit-maximizing competitive firm should increase output whenever:
- (a) Total revenue exceeds total cost.
 - (b) Average revenue exceeds average variable cost.
 - (c) Marginal revenue exceeds marginal cost.
 - (d) Price exceeds average total cost.
8. In a constant-cost industry with free entry and exit, the long-run competitive equilibrium is characterized by:
- (a) Price equal to the minimum of long-run average total cost.
 - (b) Each firm producing where marginal cost is minimized.
 - (c) Firms earning positive economic profits.
 - (d) A downward-sloping long-run supply curve.

Use the following information to answer questions 9–12. A profit-maximizing, price-taking firm produces output, widgets, y using a single input, labor, x . The wage rate of x is ₹350 per unit. The firm can produce 0, 5, 6, or 12 units of y using 0, 2, 4, or 8 units of x , respectively. There are no other input-output combinations. The firm production function is therefore given by:

$$y = f(x) = \begin{cases} 0 & \text{if } x = 0, \\ 5 & \text{if } x = 2, \\ 6 & \text{if } x = 4, \\ 12 & \text{if } x = 8. \end{cases}$$

9. What is the cost function $C(y)$ of this firm?

$$(a) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 900 & \text{if } y = 5, \\ 1,600 & \text{if } y = 6, \\ 2,800 & \text{if } y = 12. \end{cases}$$

$$(b) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 700 & \text{if } y = 5, \\ 1,400 & \text{if } y = 6, \\ 2,800 & \text{if } y = 12. \end{cases}$$

$$(c) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 1,050 & \text{if } y = 5, \\ 1,400 & \text{if } y = 6, \\ 2,800 & \text{if } y = 12. \end{cases}$$

$$(d) \ C(y) = \begin{cases} 0 & \text{if } y = 0, \\ 700 & \text{if } y = 5, \\ 1,400 & \text{if } y = 6, \\ 3,200 & \text{if } y = 12. \end{cases}$$

10. Let p be the price of a widget. What is the firm's supply function $y(p)$?

$$(a) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 140, \\ 5 & \text{if } 140 \leq p < 233.33, \\ 12 & \text{if } p \geq 233.33. \end{cases}$$

$$(b) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 116.67, \\ 5 & \text{if } 116.67 \leq p < 300, \\ 12 & \text{if } p \geq 300. \end{cases}$$

$$(c) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 140, \\ 5 & \text{if } 140 \leq p < 300, \\ 6 & \text{if } 300 \leq p < 350, \\ 12 & \text{if } p \geq 350. \end{cases}$$

$$(d) \ y(p) = \begin{cases} 0 & \text{if } 0 \leq p < 140, \\ 5 & \text{if } 140 \leq p < 300, \\ 12 & \text{if } p \geq 300. \end{cases}$$

11. Suppose that there are 6 such price taking firms. What is the industry supply $Y^S(p)$?

$$(a) \ Y^S(p) = \begin{cases} 0 & \text{if } 0 \leq p < 140, \\ 30 & \text{if } 140 \leq p < 233.33, \\ 72 & \text{if } p \geq 233.33. \end{cases}$$

$$(b) \ Y^S(p) = \begin{cases} 0 & \text{if } 0 \leq p < 116.67, \\ 30 & \text{if } 116.67 \leq p < 300, \\ 72 & \text{if } p \geq 300. \end{cases}$$

$$(c) \ Y^S(p) = \begin{cases} 0 & \text{if } 0 \leq p < 140, \\ 30 & \text{if } 140 \leq p < 300, \\ 72 & \text{if } p \geq 300. \end{cases}$$

$$(d) \ Y^S(p) = \begin{cases} 0 & \text{if } 0 \leq p < 140, \\ 30 & \text{if } 140 \leq p < 300, \\ 36 & \text{if } 300 \leq p < 350, \\ 72 & \text{if } p \geq 350. \end{cases}$$

12. Suppose the market demand function is $Y^D(p) = 170 - \frac{p}{2}$. What is the market equilibrium?
- (a) $p^* = 280$; $Y^* = 30$ units
 - (b) $p^* = 280$; $Y^* = 72$ units
 - (c) $p^* = 140$; $Y^* = 100$ units
 - (d) $p^* = 300$; $Y^* = 20$ units
13. A programmer quits a job paying \$80,000 per year to start her own firm. In the first year, her firm earns \$100,000 in revenue and incurs \$30,000 in explicit costs. Her economic profit is:
- (a) \$20,000
 - (b) -\$10,000
 - (c) \$100,000
 - (d) \$70,000
14. (Bonus) What are the names of the teaching assistants leading your tutorials?
- (a) Anel, Shynar, Daniyal, Aruzhan, Nuraily, Malika, Manat, Temirkhan, Aidana, Altynay
 - (b) Assel, Shugyla, Tantemir, Manas, Nurailym, Damir, Alima, Arailym, Merey
 - (c) Qyzbolsyn, Wifibek, Chalkabay, Qoltyqshash, Gigachad, Sahur, Uwuzhan, Qyzbolar, Paluan
 - (d) Aruzhan, Tamerlan, Malika, Shugyla, Anel, Mansur, Nurai, Daryn, Almira
 - (e) I was too cool to attend tutorials
15. The marginal rate of technical substitution (MRTS) of capital for labor is equal to:
- (a) $\frac{K}{L}$
 - (b) $\frac{AP_L}{AP_K}$
 - (c) $\frac{MP_L}{MP_K}$
 - (d) $\frac{MP_K}{MP_L}$
16. At the cost-minimizing input combination in the long run, the firm sets:
- (a) $w = r$
 - (b) $MRTS = \frac{w}{r}$
 - (c) $MP_L = MP_K$
 - (d) $\frac{MP_L}{w} = \frac{MP_K}{r}$
17. In the short run, a competitive firm should shut down if and only if:
- (a) Revenue is less than avoidable costs.
 - (b) Price is below average total cost.
 - (c) Price is below marginal cost.
 - (d) It is making negative economic profits.

18. Akmaral is the owner of a garden with apples. She wants to sell apples this summer at the local market (assume the local market is competitive). Her friend Sultan helped her calculate the short-run cost, which is

$$C(q) = k + \frac{q^3}{2} + q.$$

What is the price of shutting down?

- (a) $p = k$
- (b) $p = 1$
- (c) $p = 0$
- (d) $p = \frac{3}{2}q^2 + 1$

Consider the following two statements:

Statement 1: Consider the market for cars. All firms have the same production technology, and we know that the long-run average cost curve has a minimum at a cost of 4,600,000 Tenge per car. Therefore, we can never observe a market equilibrium in which the equilibrium price is below 4,600,000 Tenge per car.

Statement 2: Consider a firm that produces computers. Suppose that the market price of a computer is $\bar{p}$, both in the short run and the long run. If in the short run this firm decides to shut down, then it must be that in the long run it decides to exit the market.

19. Which of the following options correctly describes the truth values of the statements?

- (a) Statement 1 is **True**; Statement 2 is **False**.
- (b) Both statements are **False**.
- (c) Both statements are **True**.
- (d) Statement 1 is **False**; Statement 2 is **True**.

20. Short-run total costs are, in general, greater than or equal to long-run total costs for a given level of output because:

- (a) In the short run, at least one input is fixed, preventing the firm from choosing the cost-minimizing input combination.
- (b) Sunk costs are higher in the short run.
- (c) Input prices are higher in the short run.
- (d) Short-run production functions are different from long-run production functions.

Consider an industry that has free entry and exit. Suppose that initially it is in a long-run equilibrium with constant costs with a fixed number of firms, and zero profits. Furthermore, suppose that firms are identical, and that they have upward sloping marginal costs. With this information, answer questions 21–24.

21. In the short-run, with a fixed number of firms, the supply curve is:

- (a) Upward sloping
- (b) Horizontal and perfectly elastic
- (c) Vertical and perfectly inelastic
- (d) None of the above.

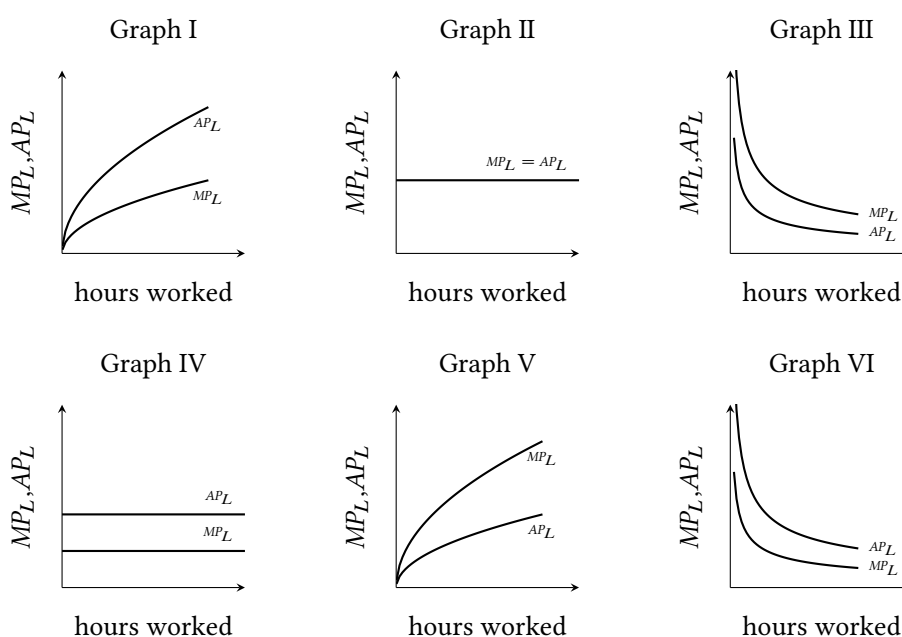
22. In the long-run, the long-run aggregate supply curve is:

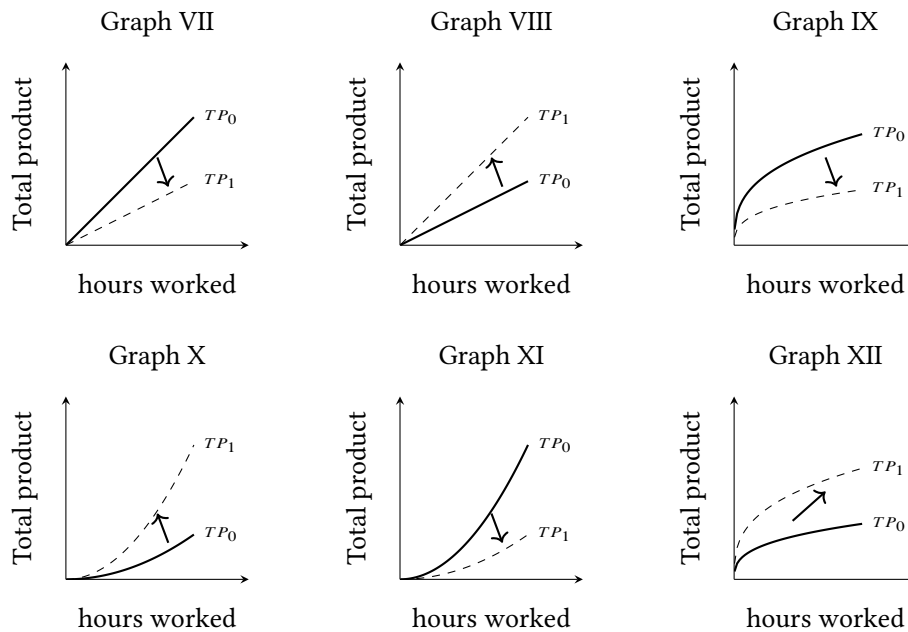
- (a) Downward sloping and equal to the minimum average cost which is decreasing.
- (b) Horizontal and equal to the minimum average cost.
- (c) Upward sloping and equal to the minimum average cost which is increasing.
- (d) I don't know and perhaps that is why I tried to pay someone to take the exam for me.

23. Now suppose that there is an increase in demand for the good that this industry produces. Then after the adjustment to the new equilibrium in the short-run:
- The new equilibrium price will be higher compared to the original equilibrium price; producers will be making economic profits.
 - The new equilibrium price will be higher compared to the original equilibrium price; producers will be making economic losses.
 - The new equilibrium price will be lower compared to the original equilibrium price; producers will be making economic profits.
 - The new equilibrium price will be lower compared to the original equilibrium price; producers will be making economic losses.
24. After the increase in the demand and the short-run adjustment, suppose that the government restricts entry into the industry by imposing a strict control on entry via operation permits. Then, in the long-run, the long-run aggregate supply is:
- Upward sloping due to the imposition of the operation permits.
 - Horizontal and equal to the minimum average cost.
 - Downward sloping due to the imposition of the operation permits.
 - I don't know because instead of studying I was trying to find someone who could take the exam for me.
25. The law of diminishing marginal productivity states that as a firm increases one input while holding all others constant, the marginal product of that input will eventually:
- Decrease
 - Remain constant
 - Increase at an increasing rate
 - Become negative immediately

Analyze the graphs below and answer questions 26 and 27.

Entrepreneur A claims that each additional worker they hire produces the same number of units as the previous worker. Entrepreneur B claims that each additional worker they hire produces fewer units than the previous worker. Entrepreneur C claims that each additional worker they hire produces more units than the immediately preceding worker.





26. The graph _____ represents the average and marginal product of Entrepreneur A. The graph _____ represents the average and marginal product of Entrepreneur B. The graph _____ represents the average and marginal product of Entrepreneur C.

- (a) II; III; I.
- (b) II; VI; V.
- (c) I; II; III.
- (d) IV; VI; V.

27. If the three companies invest in a course on worker health care and on-the-job training, the graph _____ represents Company A, the graph _____ represents Company B, and the graph _____ represents Company C.

- (a) VIII; XII; X
- (b) VI; VII; XII
- (c) VII; IX; XI
- (d) VIII; X; XII

The following information pertains to questions 28–32. Prof. Alejandro is thinking to partner with one of his past TAs, Sabina, to open a cat cafe in Astana, where customers can buy food or coffee, or just play and have fun with the cats that live there. Sabina, being the top ECON101 student of her time, understands the fundamental need to make a detailed economic analysis. It turns out that the production function of a cat cafe can be modeled as follows: Let h be the total number of happy customers served in the cafe, then we have that

$$h = F(L, K, Z) = L^{1/2} K^{1/3} Z,$$

where L is hours of labor from the cat cafe attendants, K is the number of cats in the cafe, and Z is a variable that can either be 1 or 0, and denotes whether the fixed costs of the cafe are paid (rent/lease, renovation, permits, etc.); notice that production cannot happen unless these fixed costs are paid. Sabina estimates that rent per month, plus renovation, furnishing, and monthly repair costs, plus licensing and permits would amount to a total of ₸800,000 per month, i.e., $p_Z = 800,000$. In order to determine the other costs, Sabina and Prof. Alejandro decide to hire Daniyar as a consultant, who is an authority in *creative accounting* as well as a certified cat expert. Daniyar estimates that the monthly cost of maintaining a single cat (dry food, wet food, veterinary care, grooming, insurance, basic needs, and toys) would be around ₸90,000 per cat, per month, i.e., $p_K = 90,000$.¹ Sabina and Prof. Alejandro also decide to hire Daniyar as the head attendant of the cat cafe, since he loves cats, is very good at handling

¹You may think this is very expensive, but Prof. Alejandro instructed Daniyar to estimate this number based on cats that are treated as royalty, as it should be.

difficult clients, and his career as a banker didn't pan out. They decide to pay a wage rate of ₹3,000 per hour per attendant, i.e., $w = 3,000$.

28. Suppose that Prof. Alejandro and Sabina decide to go forward, and *sink the investment of* paying the cost of Z , and start the cat cafe with $K = 10$ cats. Daniyar works 160 hours per month as the head attendant. How many customers can the cafe serve per month? You may find useful that $\sqrt{160} \approx 12.65$ and $10^{1/3} \approx 2.15$. Round your answer to the nearest integer.
- (a) $h = 22$ customers per month.
 - (b) $h = 160$ customers per month.
 - (c) $h = 34$ customers per month.
 - (d) $h = 27$ customers per month.
29. Let p_h be the average expenditure per customer. At what level of p_h is serving 27 customers per month the profit-maximizing short-run choice for the cafe? You may find useful that $10^{2/3} \approx 4.64$. Round your answer to the nearest thousand.
- (a) ₹70,000 per customer.
 - (b) ₹6,000 per customer.
 - (c) ₹35,000 per customer.
 - (d) ₹17,500 per customer.
30. Sabina also considers the long-run issue that the cafe can support, due to space, no more than $\bar{K} = 20$ cats. Assuming a target of 130 customers per month, how many hours of labor per month does the cafe need if they are housing 20 cats? You may find useful that $20^{2/3} \approx 7.37$. Round your answer to the nearest integer.
- (a) 1,147 hours per month.
 - (b) 4,588 hours per month.
 - (c) 130 hours per month.
 - (d) 2,294 hours per month.
31. From the previous problem, Prof. Alejandro and Sabina realize that Daniyar alone is not enough to run the cafe. Since the standard working time is 160 hours per month, they would need more than one attendant. Fortunately, Prof. Alejandro has an army of past TAs that are about to graduate and face unemployment, and who are willing to work for the same wage of ₹3,000 per hour. Not counting Sabina, who is a partner, how many TAs does Prof. Alejandro need to hire? Round your answer to the nearest integer.
- (a) 7 TAs.
 - (b) 1 TA, i.e., just Daniyar.
 - (c) 29 TAs.
 - (d) 14 TAs.

32. Shynar, Prof. Alejandro's former head TA and recently admitted PhD student in Economics, reviews the business plan and is not impressed. She computes the long-run average cost of serving 130 customers per month and concludes that the business idea, while charming, is not viable as stated. Which of the following best describes Shynar's concern and her suggested fix?
- The estimate of 130 customers per month implies an average expenditure of more than ₦70,000 per customer, which is unrealistically high for a cat cafe. To make the business viable, they would need significantly more customers per month, which in turn would require hiring even more of Prof. Alejandro's unemployed TAs. Fortunately, there seems to be no shortage of those.
 - The business is not viable because the ex-TA attendants are being paid too much. Shynar suggests cutting their wages, but Prof. Alejandro feels guilty given that he is the reason they are unemployed in the first place.
 - The business is not viable because the cats are not treated well enough. Shynar suggests increasing p_K , which Prof. Alejandro enthusiastically agrees with.
 - Shynar is wrong. 130 customers per month is more than enough, and ₦70,000 per customer is a perfectly reasonable price for the privilege of spending time with royal cats.

Consider the production function:

$$F(K, L) = 3L^{1/2} + K^{1/2}$$

The following information pertains to questions 33 and 34.

33. What are the returns to scale?
- Decreasing returns to scale
 - Increasing returns to scale
 - Constant returns to scale
 - Depends on K and L ratio
34. What happens to the Marginal Product of capital (MP_K) as K increases?
- Decreasing
 - First increasing, then decreasing
 - Increasing
 - Constant

Shugyla loves matcha latte from Drinkit. However, because she lives far away from the university, she cannot enjoy it every day. Nurai suggests preparing matcha latte at home using the following recipe:

- 2 teaspoons of matcha powder
- 1 cup of milk
- 2 teaspoons of hot water
- 1 teaspoon of sweetener

Let M = matcha powder, L = milk, W = hot water, S = sweetener. The following information pertains to questions 35–40.

35. According to the above recipe, what would be the appropriate production function to model the recipe that Nurai provided to Shugyla?
- $q = \frac{M + L + W + S}{4}$
 - $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S\right)$
 - $q = \min(M, L, W, S)$
 - $q = M + L + W + S$

36. Suppose Shugyla owns only one frother machine, and with one frother she can make at most 6 lattes per hour regardless of ingredients. What is the new production function?

- (a) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S, 6\right)$
- (b) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S\right) + 6$
- (c) $q = \max\left(\frac{M}{2}, L, \frac{W}{2}, S, 6\right)$
- (d) $q = 6 \min(M, L, W, S)$

37. If Shugyla has enough ingredients for 5 lattes, but only one frother, how many can she actually produce in one hour?

- (a) 6
- (b) 3
- (c) 5
- (d) 11

38. Ignoring the frother constraint from questions 36 and 37, suppose sweetener is no longer required. What is the new production function?

- (a) $q = M + L + W$
- (b) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}\right)$
- (c) $q = \frac{M}{2} + L + \frac{W}{2}$
- (d) $q = \min\left(\frac{M}{2}, L, \frac{W}{2}, S\right)$

39. Now suppose Shugyla considers only matcha powder and milk as inputs. She wants to produce 5 matcha lattes and can choose among three technologies:

- Technology 1: 5 tsp matcha powder + 5 cups milk
- Technology 2: 5 tsp matcha powder + 3 cups milk
- Technology 3: 5 tsp matcha powder + 2 cups milk

If she can freely choose which technological process to use, which verbal description below represents the correct description of the combined isoquant for producing 5 matcha lattes?

- (a) A single L-shaped isoquant at (5, 5) representing Technology 1 only.
- (b) A smooth convex curve connecting the points (5, 5), (5, 3), and (5, 2).
- (c) Three disconnected L-shaped isoquants for each technology, without any connection.
- (d) A single L-shaped isoquant at (5, 2) representing Technology 3 only.

40. Suppose milk costs 400 tenge per cup, while matcha powder costs 100 tenge per teaspoon. Which technology minimizes total cost for producing 5 lattes?

- (a) Technology 2
- (b) Technology 1
- (c) Technology 3
- (d) All technologies have the same cost